Literature Review: Brazilian E-Commerce Public Dataset

Origins of e-Commerce

The sale of goods and services from any seller to any buyer across the globe through purely electronic means, now referred to as “electronic commerce”, a.k.a. “e-commerce”, had its origins with the founding of CompuServe in 1969.[1] CompuServe began as a computer resource time-sharing service, and in 1979 was the first company to provide email services to individual, private users (“personal computer users”). By following the next year (1980) with introduction of real-time, online chat between users, CompuServe enabled the exchange of information instantly and a means for private citizens to advertise and communicate both publicly and privately without the use of a telephone or printed media[1], for example classified ads in newspapers or personal ad sections in various magazines. In parallel to CompuServe’s development, connecting computers through data lines to eventually enable e-commerce began with the connection of the first two nodes of the Advanced Research Projects Agency Network (ARPANet) at UCLA and Stanford, also in 1969.[2] Remote connectivity to ARPANet’s nodes began two years later in 1971 through direct dialed connections, and connectivity expanded outside US borders with the connection of a node at University College London to ARPANet via a combination of satellite and cable connections in 1973.[3] Once international connectivity was established, the next biggest barrier to e-commerce was security concerns with online payment, which took decades to overcome. The first fully online transaction (including payment) didn’t come until 1994, and is credited either to Dan Kohn, creator of the NetMarket web market, or to the Internet Shopping Network company through their website, both transactions occurring somewhere between July and August of that year.[4] No matter whose claim is correct, in the 20 years since, global e-commerce revenues have grown from those simple, small transactions to approximately US$6 trillion worldwide in 2024.[5]

E-Commerce and Analytics

Now that e-commerce is big business, and many companies have a huge amount of transaction data to analyze, success in e-commerce is driven by answering important business questions with that data and taking appropriate actions. Those questions concern the same categories important to businesses regardless of where or what they're selling:[6]

- Customers – demographics and demographic trends/groups, customer preferences, buying trends, what makes them buy more and more often
- Products/services – what drives sales, why customers buy them, who is buying what and where
- Logistics – Are shipments on time, what drives late deliveries, what carriers are consistently delinquent, where are bottlenecks occurring

These categories of interest drive the questions to be investigated in this project, as well.

Project Dataset and Questions for Investigation

Olist, one of the largest department stores in Brazil, hosts an online store that offers not only products from Olist, but also from small businesses throughout Brazil via a standard contract, with shipping handled directly by Olist via their logistics network.[7] They have provided a dataset (via Kaggle) consisting of over 100k transactions across 18 months of sales (12/31/16 – 5/31/18) with details concerning customer location (including geolocation data), order size and ordered products, seller information, product category and description information, and payment and delivery information. For this project, the authors intend to investigate the answers to the following business questions:

1. What factors contribute to late deliveries, and how accurately can we predict them in advance?
2. Are there any sellers that are consistently delinquent delivering orders to logistics carriers, and does late delivery to the carrier reliably predict late delivery to the customer? Within those consistently late sellers, if any, are there any significant sub-factors that correlate with that lateness?
3. Do different customer segments (i.e. new/repeat customers, location groupings, age groupings) show significant differences in order cancellation and return rates? Can these behaviors be predicted using early indicators such as browsing time or items added to cart but not purchased?

Due to the age of the dataset and it being hosted on a public forum for data analytics, there's been a significant amount of investigation already performed directly on it. The most relevant (for this project) and comprehensive look is also the most recent, having been posted 1 month ago by Sergei Egorov, a student at Tomsk State University in Russia, though his investigation only included 22 of the 52 available fields (12 of the 52 being key fields). His analysis showed that the late delivery rate across all transactions was 6.5%, which is relevant to question 1, but did not dig into the reasons for late deliveries. Most of his analysis tried to tie the metrics associated with delivery to customer satisfaction/ratings and follow-on business, and so does not overlap with the questions for this project.[8] No other code submissions appear to have analyzed delivery performance in any ways relevant to questions 1 and 2, with the only submission that stated it would predict delivery delay having created an allegedly optimized model, but failed to state any conclusions or even address the accuracy of the alleged best-fit, optimized model.[9] There does not appear to have been any significant analysis directed towards order cancellations and product returns, with the only submission directly related to either stating only that there were 553 orders cancelled across the >115k total orders in the dataset. That small a sample size for analysis may prohibitively affect the ability to answer question 3.

References

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